

Topics for week 5

Chapter 3: Linear Transformation, Eigenvalues and Eigenvectors

3.5 Diagonalization

3.6 Powers of Matrices

Chapter 4: System of Differential Equations

4.1 System of Differential Equations

4.2 Solutions of SDE

4.3 Phase Plane

Why do Engineers care about eigenvalues & eigenvectors?

Finding stability:

Eigenvalue $\lambda = 1$

$$\mathbf{A}\mathbf{v} = \mathbf{v} \text{ (no change)} \Rightarrow \mathbf{A}^2\mathbf{v} = \mathbf{v} \Rightarrow \mathbf{A}^3\mathbf{v} = \mathbf{v} \dots \Rightarrow \mathbf{A}^n\mathbf{v} = \mathbf{v}$$

Steady state

Eigenvalue $\lambda = 2$

$$\mathbf{A}\mathbf{v} = 2\mathbf{v} \text{ (2 times)} \Rightarrow \mathbf{A}(2\mathbf{v}) = 2(2\mathbf{v}) \text{ (2 times)} \Rightarrow \mathbf{A}(4\mathbf{v}) = 2(4\mathbf{v}) \text{ (2 times)}$$

Steady growth

$$\mathbf{A} \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 10 \end{pmatrix}$$

$$\mathbf{A} \begin{pmatrix} 4 \\ 6 \\ 10 \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \\ 20 \end{pmatrix}$$

$$\mathbf{A} \begin{pmatrix} 8 \\ 12 \\ 20 \end{pmatrix} = \begin{pmatrix} 16 \\ 24 \\ 40 \end{pmatrix}$$

Input/output vectors maintain the same component ratio 2 : 3 : 5

Steady proportion

$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} \longrightarrow \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \ddots & & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_k \end{pmatrix}$

diagonalizable matrix diagonal matrix

Diagonalization

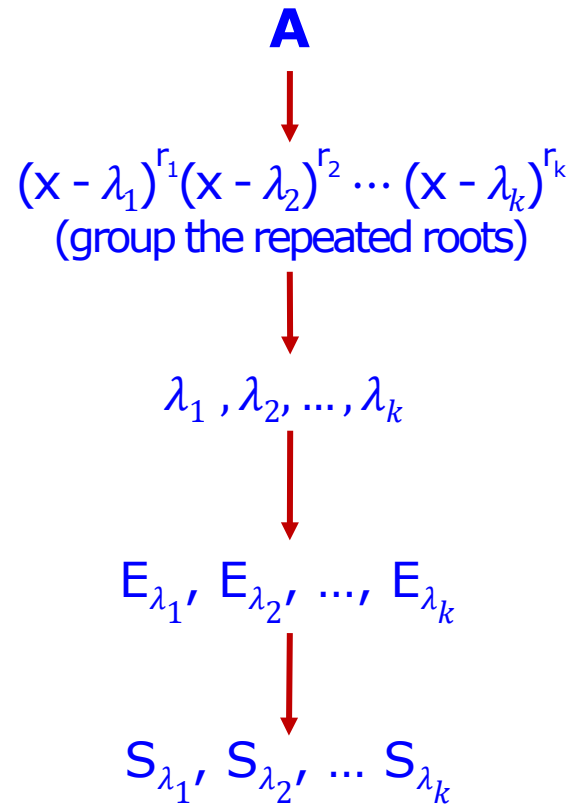
$n \times n$ matrix

characteristic polynomial

eigenvalues

eigenspaces

bases for eigenspaces



Case 1: $\dim E_{\lambda_i} = r_i$
for all i

$\dim E_{\lambda_1} + \dim E_{\lambda_2} + \dots + \dim E_{\lambda_k} = n$
 $S_{\lambda_1} \cup S_{\lambda_2} \cup \dots \cup S_{\lambda_k}$
 contains **exactly** n
 linearly independent
 eigenvectors
 $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$

form $n \times n$ matrix
 $\mathbf{P} = (\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_n)$

A diagonalizable

Case 2: $\dim E_{\lambda_i} < r_i$
for some i

$\dim E_{\lambda_1} + \dim E_{\lambda_2} + \dots + \dim E_{\lambda_k} < n$
 $S_{\lambda_1} \cup S_{\lambda_2} \cup \dots \cup S_{\lambda_k}$
 contains **less than** n
 linearly independent
 eigenvectors
 $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$

cannot form
 $n \times n$ matrix **P**

A not diagonalizable

Example 1: Diagonalization

$$\mathbf{A} = \begin{pmatrix} 3 & -2 & -1 \\ 0 & 5 & 1 \\ 0 & 2 & 4 \end{pmatrix} \quad \text{Characteristic polynomial: } (x - 3)^2(x - 6)$$

eigenvalues: 3 (repeated), 6

Diagonalize $\mathbf{A}$

eigenvectors associated to 3: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}$

eigenvectors associated to 6: $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

by right: $\begin{pmatrix} 0 \\ -1/2 \\ 1 \end{pmatrix}$

3 linearly independent
eigenvectors
 $\Rightarrow \mathbf{A}$ is diagonalizable

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 2 & 1 \end{pmatrix}$$

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

$$\mathbf{R} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\mathbf{R}^{-1}\mathbf{A}\mathbf{R} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\mathbf{Q} = \begin{pmatrix} 5 & 0 & -2 \\ 0 & 1 & 2 \\ 0 & -2 & 2 \end{pmatrix}$$

$$\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

$$\mathbf{S} = \begin{pmatrix} 1 & 1 & 3 \\ -1 & 1 & -3 \\ 2 & -2 & -3 \end{pmatrix}$$

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

If an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable.

scalar matrix: $\begin{pmatrix} a & 0 & \dots & 0 \\ 0 & a & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a \end{pmatrix}$

Example 2: Triangular Matrices

Are triangular matrices diagonalizable?

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix}$$

3 distinct eigenvalues
diagonalizable

$$\dim E_1 = 1, \dim E_2 = 1, \dim E_3 = 1$$

$$C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

only 2 eigenvalues
diagonalizable
can't tell by inspection

$$\dim E_1 = 2, \dim E_2 = 1$$

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

only 1 eigenvalue
B not scalar matrix
non-diagonalizable

$$\dim E_1 = 1$$

$$D = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

only 2 eigenvalues
non-diagonalizable
can't tell by inspection

$$\dim E_1 = 1, \dim E_2 = 1$$

matrix with
exactly 1 eigenvalue

scalar matrix
diagonalizable

not scalar matrix
not diagonalizable

Find the dimension of E_1

Q&A: log in to [PollEv.com/vtpoll](https://www.pollen.com/vtpoll)

Example 3:

Power of Diagonalizable Matrices

$$\mathbf{A} = \begin{pmatrix} 3 & -2 & -1 \\ 0 & 5 & 1 \\ 0 & 2 & 4 \end{pmatrix}$$

Find $\mathbf{A}^5$

from eigenvectors

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 2 & 1 \end{pmatrix}$$

from eigenvalues

$$\mathbf{D} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

direct compute

$$\mathbf{P}^{-1} = \begin{pmatrix} 1 & 2/3 & 1/3 \\ 0 & -1/3 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix}$$

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix} \xrightarrow{\text{red arrow}} \mathbf{A} = \mathbf{P} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{pmatrix} \mathbf{P}^{-1} \xrightarrow{\text{red arrow}} \mathbf{A}^5 = \mathbf{P} \begin{pmatrix} 3^5 & 0 & 0 \\ 0 & 3^5 & 0 \\ 0 & 0 & 6^5 \end{pmatrix} \mathbf{P}^{-1}$$

$$\mathbf{A}^5 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 243 & 0 & 0 \\ 0 & 243 & 0 \\ 0 & 0 & 7776 \end{pmatrix} \begin{pmatrix} 1 & 2/3 & 1/3 \\ 0 & -1/3 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix} = \begin{pmatrix} 243 & -5022 & -2511 \\ 0 & 5265 & 2511 \\ 0 & 5022 & 2754 \end{pmatrix}$$

Example 4: Population Model

In a certain town, 30% of married women get divorced each year and 20% of women get married each year.

There are 8000 married women and 2000 single women.

What's the population of married and single women in the long run?

M_n = married women after n years

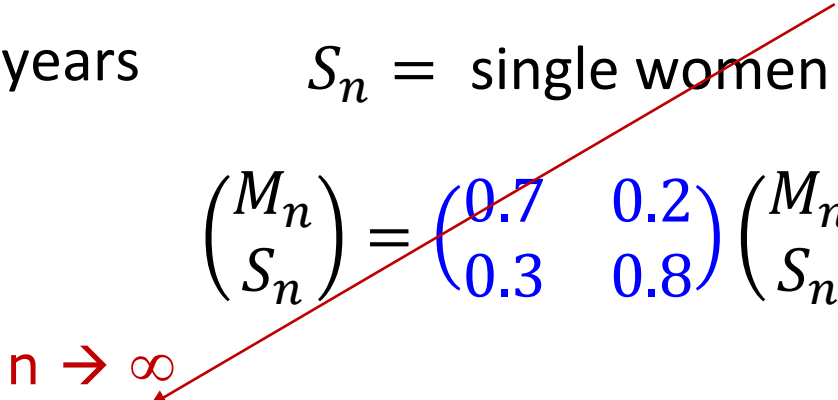
S_n = single women after n years

$$M_n = 0.7M_{n-1} + 0.2S_{n-1}$$

$$S_n = 0.3M_{n-1} + 0.8S_{n-1}$$

$$\begin{pmatrix} M_n \\ S_n \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix} \begin{pmatrix} M_{n-1} \\ S_{n-1} \end{pmatrix}$$

After n year $\begin{pmatrix} M_n \\ S_n \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^n \begin{pmatrix} 8000 \\ 2000 \end{pmatrix}$



What's the population of married and single women in the long run?

Example 4: Population Model (cont)

In a certain town, 30% of married women get divorced each year and 20% of women get married each year.

There are 8000 married women and 2000 single women.

Diagonalise

$$\begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}$$

eigenvalues: 1 and 0.5

eigenvectors associated to 1: $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$,

eigenvectors associated to 0.5: $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$$\begin{pmatrix} M_n \\ S_n \end{pmatrix} = \begin{pmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{pmatrix}^n \begin{pmatrix} 8000 \\ 2000 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0.5^n \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix}^{-1}} \begin{pmatrix} 8000 \\ 2000 \end{pmatrix}$$

Expand this product to find M_n and S_n in terms of n

$$\text{In the long run} \rightarrow \begin{pmatrix} 2 & -1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/5 & 1/5 \\ -3/5 & 2/5 \end{pmatrix} \begin{pmatrix} 8000 \\ 2000 \end{pmatrix} = \begin{pmatrix} 0.4 & 0.4 \\ 0.6 & 0.6 \end{pmatrix} \begin{pmatrix} 8000 \\ 2000 \end{pmatrix} = \begin{pmatrix} 4000 \\ 6000 \end{pmatrix}$$

System of Differential Equations (SDE)

A (single) differential equation: $y'(t) = 2y(t)$

$y(t)$ is a (variable) function

$y'(t)$ is the derivative of $y(t)$

A solution is the function: $y(t) = e^{2t}$

A system of differential equations:

$$\begin{cases} y_1'(t) &= y_1(t) + 2y_2(t) \\ y_2'(t) &= -y_1(t) + 4y_2(t) \end{cases}$$

$y_1(t), y_2(t)$ are (variable) functions

A solution is the pair of functions: $y_1(t) = 2e^{2t} + e^{3t}$, $y_2(t) = e^{2t} + e^{3t}$

Check: $y_1'(t) = 4e^{2t} + 3e^{3t} = y_1(t) + 2y_2(t)$

$$y_2'(t) = 2e^{2t} + 3e^{3t} = -y_1(t) + 4y_2(t)$$

Expressing SDE

SDE in standard form

2 x 2 SDE $\begin{cases} y_1'(t) = y_1(t) + 2y_2(t) \\ y_2'(t) = -y_1(t) + 4y_2(t) \end{cases}$

SDE (with t hidden)

$$\begin{cases} y_1' = y_1 + 2y_2 \\ y_2' = -y_1 + 4y_2 \end{cases}$$

a solution

$$\begin{aligned} y_1(t) &= 2e^{2t} + e^{3t} \\ y_2(t) &= e^{2t} + e^{3t} \end{aligned}$$

SDE in matrix form

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$\mathbf{y}' = \mathbf{A}\mathbf{y}$

a solution in matrix form

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 2e^{2t} + e^{3t} \\ e^{2t} + e^{3t} \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}$$

underlying matrix
of the SDE

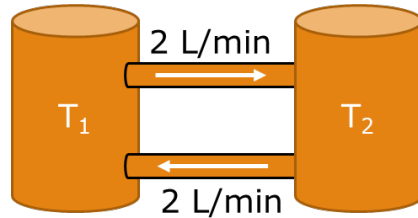
Can be extended to n x n SDE

Amounts of fertilizer at time t :
Tank T_1 : $y_1(t)$ Tank T_2 : $y_2(t)$

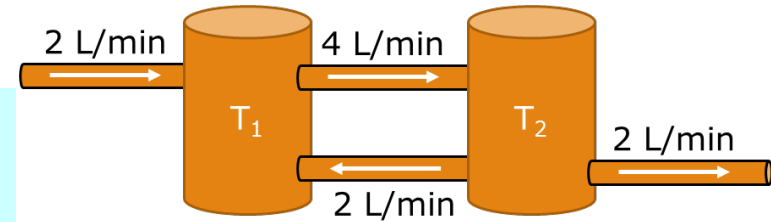
Rate of change of fertilizer at time t :
Tank T_1 : $y_1'(t)$ Tank T_2 : $y_2'(t)$

Example 5: Mixing Problem of two tanks

Tank T_1 and T_2 contain 100L of water each, with certain amount of fertilizer



rate of change
= Inflow/min – Outflow/min



$$\text{Tank } T_1: y_1'(t) = \frac{2}{100} y_2(t) - \frac{2}{100} y_1(t)$$

$$\text{Tank } T_2: y_2'(t) = \frac{2}{100} y_1(t) - \frac{2}{100} y_2(t)$$

$$y_1'(t) = -0.02 y_1(t) + 0.02 y_2(t)$$

$$y_2'(t) = 0.02 y_1(t) - 0.02 y_2(t)$$

$$\text{Tank } T_1: y_1'(t) = 0 + \frac{2}{100} y_2(t) - \frac{4}{100} y_1(t)$$

$$\text{Tank } T_2: y_2'(t) = \frac{4}{100} y_1(t) - \frac{2}{100} y_2(t) - \frac{2}{100} y_2(t)$$

$$y_1'(t) = -0.04 y_1(t) + 0.02 y_2(t)$$

$$y_2'(t) = 0.04 y_1(t) - 0.04 y_2(t)$$

→ **A** is diagonalizable

Solutions of $\mathbf{y}' = \mathbf{A}\mathbf{y}$

If $\mathbf{v}$ is an eigenvector of an $n \times n$ matrix **A** with eigenvalue r :

Then $\mathbf{y} = \mathbf{v}e^{rt}$ is a solution of the SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$

Suppose **A** has n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$
with eigenvalues $r_1, r_2, \dots, r_n$.

$\mathbf{v}_1e^{r_1t}, \mathbf{v}_2e^{r_2t}, \dots, \mathbf{v}_ne^{r_nt}$ are n linearly independent solutions of the SDE

$\{\mathbf{v}_1e^{r_1t}, \mathbf{v}_2e^{r_2t}, \dots, \mathbf{v}_ne^{r_nt}\}$ forms a "basis" for the solution space of the SDE
a fundamental set of solutions

$\mathbf{y} = c_1\mathbf{v}_1e^{r_1t} + c_2\mathbf{v}_2e^{r_2t} + \dots + c_n\mathbf{v}_ne^{r_nt}$ gives the general solution of the SDE

$\mathbf{v}_1 e^{r_1 t}, \mathbf{v}_2 e^{r_2 t}, \dots, \mathbf{v}_n e^{r_n t}$ are n linearly independent solutions of the SDE

Example 6: General solution

$$\begin{cases} y_1' &= 3y_1(t) - 2y_2(t) - y_3(t) \\ y_2' &= 5y_2(t) + y_3(t) \\ y_3' &= 2y_2(t) + 4y_3(t) \end{cases} \quad \begin{pmatrix} 3 & -2 & -1 \\ 0 & 5 & 1 \\ 0 & 2 & 4 \end{pmatrix}$$

eigenvalues: 3 (repeated), 6
eigenvectors associated to 3: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}$
eigenvectors associated to 6: $\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{3t}, \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} e^{3t}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} e^{6t}$ 3 linearly independent solutions of the SDE
a fundamental set of solution for the SDE

general solution of the SDE

$$\begin{pmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{3t} + C_2 \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} e^{3t} + C_3 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} e^{6t} = \begin{pmatrix} C_1 e^{3t} - C_3 e^{6t} \\ -C_2 e^{3t} + C_3 e^{6t} \\ 2C_2 e^{3t} + C_3 e^{6t} \end{pmatrix}$$

$$y_1(t) = C_1 e^{3t} - C_3 e^{6t}$$

$$y_2(t) = -C_2 e^{3t} + C_3 e^{6t}$$

$$y_3(t) = 2C_2 e^{3t} + C_3 e^{6t}$$

Example

Given the 2 x 2 SDE $\mathbf{y}' = \mathbf{A}\mathbf{y}$

Suppose $\mathbf{A}$ has two eigenvectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ with eigenvalues 1 and 2 resp.

Which of the followings are solutions of the SDE?

$\begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t$ Yes. $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is an eigenvector with eigenvalue 1

$\begin{pmatrix} 4 \\ 2 \end{pmatrix} e^{2t}$ Yes. $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ is an eigenvector with eigenvalue 2

$\begin{pmatrix} 2 \\ 1 \end{pmatrix} e^t + \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{2t}$ No. $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is not an eigenvector with eigenvalue 1,
and $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is not an eigenvector with eigenvalue 2

$\begin{pmatrix} 2 \\ 4 \end{pmatrix} e^t - \begin{pmatrix} 6 \\ 3 \end{pmatrix} e^{2t}$ Yes. $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ is an eigenvector with eigenvalue 1,
and $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$ is an eigenvector with eigenvalue 2

Example 7: Initial Value Problem

$$\begin{cases} y_1' &= 3y_1(t) - 2y_2(t) - y_3(t) \\ y_2' &= 5y_2(t) + y_3(t) \\ y_3' &= 2y_2(t) + 4y_3(t) \end{cases}$$

$$\begin{pmatrix} y_1(0) \\ y_2(0) \\ y_3(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} \quad \text{initial condition (when } t = 0 \text{)}$$

general solution: $\begin{pmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} e^{3t} + c_3 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} e^{6t}$

Substituting $t = 0$: $\begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} + c_3 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$

$$\begin{aligned} c_1 - c_3 &= 1 \\ -c_2 + c_3 &= 1 \\ 2c_2 + c_3 &= 4 \end{aligned}$$

particular solution for the IVP: $\begin{pmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{3t} + 1 \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix} e^{3t} + 2 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} e^{6t} \quad c_1 = 3, c_2 = 1, c_3 = 2$

$$y_1(t) = 3e^{3t} - 2e^{6t}$$

$$y_2(t) = -e^{3t} + 2e^{6t}$$

$$y_3(t) = 2e^{3t} + 2e^{6t}$$

Initial Value Problem

Initial value problem (IVP)

SDE
 $\mathbf{y}' = \mathbf{A} \mathbf{y}$



initial condition
 $\mathbf{y}(t_0) = \mathbf{v}$

general solution

substitute $t = t_0$

Become a
linear system

$$\mathbf{y}(t_0) = c_1 \mathbf{v}_1 e^{r_1 t_0} + c_2 \mathbf{v}_2 e^{r_2 t_0} + \dots + c_n \mathbf{v}_n e^{r_n t_0}$$

Solve for specific values:

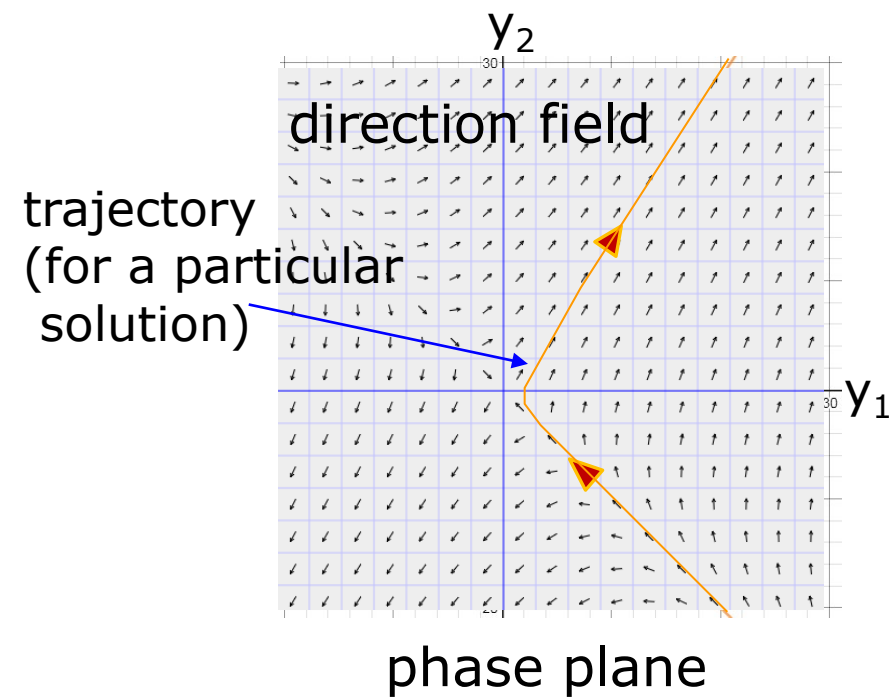
$$c_1 = a_1, c_2 = a_2, \dots, c_n = a_n$$

Particular solution (unique)

$$\mathbf{y}_p = a_1 \mathbf{v}_1 e^{r_1 t} + a_2 \mathbf{v}_2 e^{r_2 t} + \dots + a_n \mathbf{v}_n e^{r_n t}$$

This solution is unique

Graph of Solutions of $\mathbf{y}' = \mathbf{A}\mathbf{y}$



Trajectory

t	...	-2	-1	0	1	2	...
y_1		10	5	2	5	8	
y_2		-10	-5	0	5	10	

Every trajectory has a direction (move from negative t to positive t)

Particular solution

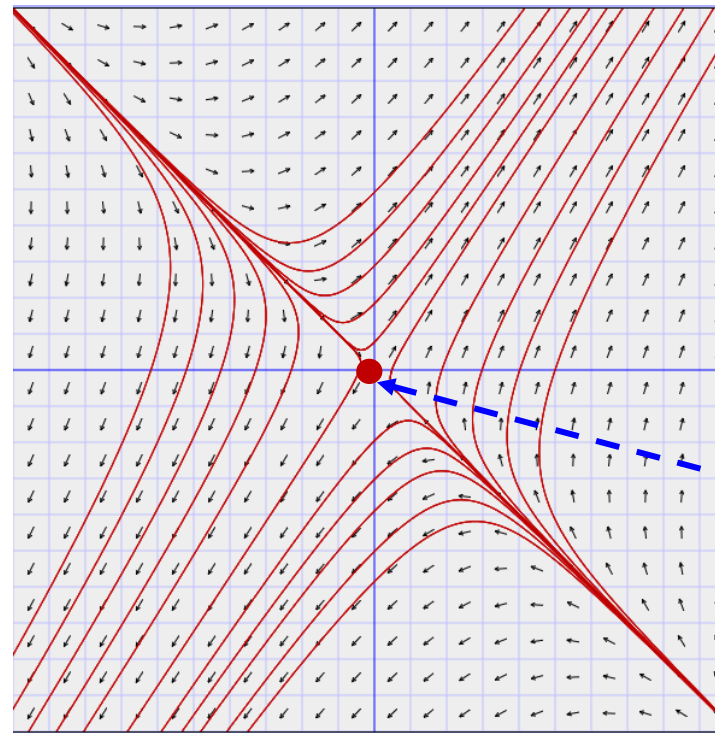
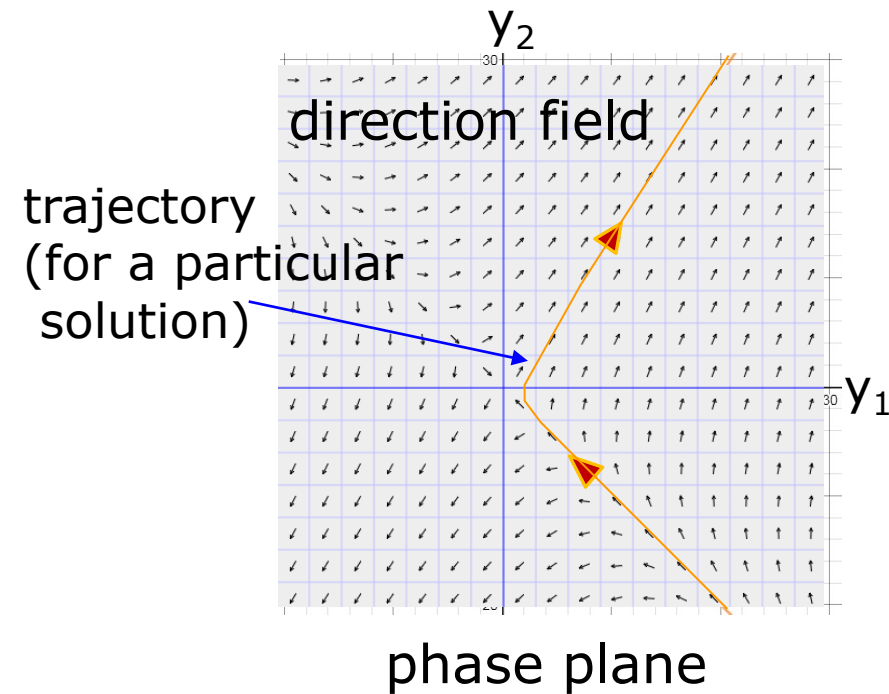
Example: Take $c_1 = c_2 = 1$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 1e^{r_1 t} \begin{pmatrix} p \\ q \end{pmatrix} + 1e^{r_2 t} \begin{pmatrix} r \\ s \end{pmatrix}$$

Direction field

$\mathbf{y}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 3 \\ 1 \end{pmatrix}$	...	...
$\mathbf{y}' = \mathbf{A}\mathbf{y}$	$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 4 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 4 \end{pmatrix}$	...	...

Phase Portrait of $\mathbf{y}' = \mathbf{A}\mathbf{y}$



phase portrait

Particular solution

Example: Take $c_1 = c_2 = 1$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 1e^{r_1 t} \begin{pmatrix} p \\ q \end{pmatrix} + 1e^{r_2 t} \begin{pmatrix} r \\ s \end{pmatrix}$$

Repeat for other pairs of c_1, c_2

when $c_1 = 0, c_2 = 0$
equilibrium solution

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Example 9: Converting ODE to SDE

2nd order ODE (Ordinary Differential Equation)

$$2y'' - 5y' + y = 0$$

1 equation with
1 variable y (function)

first order linear SDE

$$\begin{cases} y_1' = y_2 \\ y_2' = -\frac{1}{2}y_1 + \frac{5}{2}y_2 \end{cases}$$

2 equations with
2 variables y_1 and y_2

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{1}{2} & \frac{5}{2} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

- Introduce variables y_1 and y_2
 $y_1 = y$
 $y_2 = y'$
- Express y_1' and y_2' in terms of y_1 and y_2

$$ay'' + by' + cy = 0$$

$$\begin{cases} y_1' = y_2 \\ y_2' = -\frac{c}{a}y_1 - \frac{b}{a}y_2 \end{cases}$$

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$